

Stigmergic Attention: Closing the Feedback Loop Between Pheromone Fields and Neural Attention

Research Notes

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Abstract

We introduce *stigmergic attention*, a modification of the transformer attention mechanism that augments standard query-key matching with a decaying pheromone field—an additive bias accumulated from past deposits and subject to uniform erosion. Drawing on the theory of phase transitions in lattice ant models, we identify a precise three-ingredient recipe for edge-of-chaos dynamics: *amplification* (positive feedback between field and output), *injection* (content-dependent deposits uncorrelated with the field), and *erosion* (uniform exponential decay). We show that a naïve implementation lacking positive feedback reduces to an inert decaying average with no phase transition, while closing the feedback loop—computing deposits from attention *output* rather than input—recovers the full stigmergic dynamics. We present the architecture, its connection to linear state-space models, and experimental results on in-context learning tasks.

1 Introduction

Stigmergy is the mechanism by which organisms coordinate through modification of a shared environment rather than direct communication. The canonical example is ant trail formation: ants deposit pheromone as they walk, subsequent ants preferentially follow high-pheromone paths, and pheromone evaporates uniformly over time. This creates a positive feedback loop that, at the right balance between deposition and evaporation, produces emergent structure at the edge of chaos [1].

Transformers [2] coordinate information flow through attention—a content-based routing mechanism where each position queries all others. We ask: can the stigmergic mechanism augment attention with a complementary structural channel? Specifically, can a decaying deposit field, accumulated causally across sequence positions, provide useful inductive bias for in-context learning?

Our investigation proceeds in three stages:

1. We define stigmergic attention as standard attention plus an additive field bias from a linear recurrence (Section 2).
2. We identify, via the ant trail phase transition literature, that positive feedback is the critical missing ingredient, and show how to close the loop (Section 3).
3. We present experiments comparing feedback and non-feedback variants across evaporation rates (Section 5).

2 Architecture

2.1 Standard Attention

Given an input sequence $X \in \mathbb{R}^{T \times d}$, standard causal multi-head attention computes, for each head $h \in \{1, \dots, H\}$:

Definition 1 (Scaled dot-product attention). Let $\mathbf{q}_i, \mathbf{k}_j \in \mathbb{R}^{d_h}$ be the query and key vectors at positions i and j respectively, with $d_h = d/H$ the head dimension. The attention weight from position i to position j is:

$$a_{ij}^{(h)} = \frac{\exp(\mathbf{q}_i^{(h)\top} \mathbf{k}_j^{(h)} / \sqrt{d_h})}{\sum_{k \leq i} \exp(\mathbf{q}_i^{(h)\top} \mathbf{k}_k^{(h)} / \sqrt{d_h})} \quad \text{for } j \leq i \quad (1)$$

where the restriction $j \leq i$ enforces causal masking. The output at position i is:

$$\mathbf{o}_i = \text{Concat}_{h=1}^H \left(\sum_{j \leq i} a_{ij}^{(h)} \mathbf{v}_j^{(h)} \right) \cdot W_O \quad (2)$$

with queries $\mathbf{q}^{(h)} = XW_Q^{(h)}$, keys $\mathbf{k}^{(h)} = XW_K^{(h)}$, values $\mathbf{v}^{(h)} = XW_V^{(h)}$, and output projection $W_O \in \mathbb{R}^{d \times d}$.

2.2 The Pheromone Field

We augment attention with a scalar field per head that accumulates deposits and decays over positions.

Definition 2 (Pheromone field). For head h , the pheromone field at position i is defined by the linear recurrence:

$$\phi_i^{(h)} = \alpha \cdot \phi_{i-1}^{(h)} + d_i^{(h)}, \quad \phi_0^{(h)} = d_0^{(h)} \quad (3)$$

where:

- $\alpha = 1 - \varepsilon \in (0, 1)$ is the retention rate, with ε the evaporation rate
- $d_i^{(h)} \in \mathbb{R}$ is the deposit at position i for head h
- The field has initial condition $\phi_{-1}^{(h)} = 0$

Proposition 1 (Closed-form solution). The recurrence in Equation (3) has the closed-form solution:

$$\phi_i^{(h)} = \sum_{j=0}^i \alpha^{i-j} \cdot d_j^{(h)} \quad (4)$$

which can be computed as a matrix-vector product $\boldsymbol{\phi}^{(h)} = D\mathbf{d}^{(h)}$ where $D \in \mathbb{R}^{T \times T}$ is the lower-triangular decay matrix with entries $D_{ij} = \alpha^{i-j}$ for $j \leq i$ and 0 otherwise.

Proof. By induction. Base case: $\phi_0 = d_0 = \alpha^0 d_0$. Inductive step: $\phi_i = \alpha \sum_{j=0}^{i-1} \alpha^{i-1-j} d_j + d_i = \sum_{j=0}^{i-1} \alpha^{i-j} d_j + \alpha^0 d_i = \sum_{j=0}^i \alpha^{i-j} d_j$. $\square$

Remark 1 (Connection to state-space models). Equation (3) is a first-order linear recurrence—the simplest possible state-space model (SSM). It is a scalar, per-head instance of the general form $s_i = As_{i-1} + Bu_i$ with $A = \alpha \in \mathbb{R}$ and $B = 1$. Unlike $S4$ or Mamba, which use structured matrices $A \in \mathbb{R}^{n \times n}$ for multi-dimensional state, the pheromone field uses a single scalar state per head, making it both interpretable and cheap.

2.3 Stigmergic Attention

The field enters the attention computation as an additive bias on the pre-softmax logits.

Definition 3 (Stigmergic attention). *The stigmergic attention logit from position i to position j is:*

$$\ell_{ij}^{(h)} = \frac{\mathbf{q}_i^{(h)\top} \mathbf{k}_j^{(h)}}{\sqrt{d_h}} + \phi_j^{(h)} \quad (5)$$

The attention weights are obtained by applying causal masking and softmax over the logit vector $\ell_i^{(h)} = [\ell_{i1}^{(h)}, \dots, \ell_{iT}^{(h)}]$ as in Definition 1.

The field bias $\phi_j^{(h)}$ is *position-indexed*: it biases attention *toward* position j across all queries $i \geq j$. Positions with high accumulated deposits attract more attention, analogous to how pheromone-rich trail segments attract more ants.

The key design question is: where do deposits come from?

3 Theory: The Three-Ingredient Recipe

3.1 Lessons from Ant Trail Phase Transitions

The lattice ant model [1] studies pheromone trail formation on a discrete grid. The continuum limit of the pheromone dynamics is:

$$\frac{\partial \phi}{\partial t} = \underbrace{\alpha_{\text{inj}}}_{\text{injection}} + \underbrace{\beta \cdot g(\phi)}_{\text{amplification}} - \underbrace{\lambda \cdot \phi}_{\text{erosion}} + \underbrace{D \nabla^2 \phi}_{\text{diffusion}} \quad (6)$$

The paper’s central finding is that edge-of-chaos dynamics—the boundary between frozen (ordered) and disordered phases—requires exactly three ingredients, all *structurally uncorrelated* with the field ϕ :

1. **Amplification:** Positive feedback where the field influences behavior, which in turn reinforces the field. In ants: high-pheromone trails attract more ants, who deposit more pheromone.
2. **Injection:** An unconditional source of deposits that is uncorrelated with the current field state. In ants: all ants deposit pheromone regardless of the existing trail strength.
3. **Erosion:** Uniform removal that is uncorrelated with the field state. In ants: pheromone evaporates at a constant rate everywhere.

The paper tested seven alternative mechanisms that replace uniform erosion with state-dependent processes (habituation, peaked response functions, Hill functions, coupled deposition). *Every alternative failed* to produce edge-of-chaos dynamics, because state-dependent disorder is correlated with the field—it preferentially targets the structures that contribute most to the order parameter.

3.2 Mapping to Stigmergic Attention

We now map the three ingredients to the neural architecture:

The critical question is whether the architecture implements *amplification*—positive feedback between the field and the deposit process.

Ingredient	Ant system	Neural analogue	Property
Amplification	Ants follow pheromone, deposit more on trails	Field biases attention, output generates deposits	Positive feedback (ϕ -correlated)
Injection	All ants deposit unconditionally	Token content produces deposits via W_d	Uncorrelated with ϕ
Erosion	Uniform evaporation at rate λ	Exponential decay at rate ε per position	Uncorrelated with ϕ

Table 1: Three-ingredient mapping from ant systems to stigmergic attention.

3.3 The Non-Feedback Architecture (Negative Result)

In the naïve implementation, deposits are computed from the *input* to the attention layer:

$$d_i^{(h)} = W_d^{(h)} \mathbf{x}_i \quad (7)$$

where $W_d^{(h)} \in \mathbb{R}^{1 \times d}$ is a learned projection and $\mathbf{x}_i \in \mathbb{R}^d$ is the input embedding at position i .

Proposition 2 (No positive feedback without output deposits). *When deposits are computed from inputs via Equation (7), the deposit $d_i^{(h)}$ is independent of the field $\phi_0^{(h)}, \dots, \phi_{i-1}^{(h)}$. The field is therefore a deterministic function of the input sequence alone:*

$$\phi_i^{(h)} = \sum_{j=0}^i \alpha^{i-j} \cdot W_d^{(h)} \mathbf{x}_j \quad (8)$$

This is a causally-weighted moving average of input projections—an exponentially-weighted positional encoding with no feedback mechanism.

Without amplification, the field cannot create or maintain structure beyond what the input sequence provides. No phase transition is possible because there is no mechanism for self-reinforcement.

3.4 Closing the Feedback Loop

The fix is to compute deposits from the attention *output*:

Definition 4 (Feedback stigmergic attention). *At each position i , the computation proceeds causally:*

1. Compute field bias from accumulated past deposits:

$$\phi_i^{(h)} = \alpha \cdot \phi_{i-1}^{(h)} + d_{i-1}^{(h)} \quad (9)$$

2. Compute attention weights with field bias:

$$a_{ij}^{(h)} = \text{softmax}_j \left(\frac{\mathbf{q}_i^{(h)\top} \mathbf{k}_j^{(h)}}{\sqrt{d_h}} + \phi_j^{(h)} \right) \quad \text{for } j \leq i \quad (10)$$

3. Compute attention output at position i :

$$\mathbf{o}_i^{(h)} = \sum_{j \leq i} a_{ij}^{(h)} \mathbf{v}_j^{(h)} \quad (11)$$

4. Compute deposit from output:

$$d_i^{(h)} = W_d^{(h)} \mathbf{o}_i \quad (12)$$

where $\mathbf{o}_i = \text{Concat}_h(\mathbf{o}_i^{(h)}) \cdot W_O$.

This creates the closed feedback loop:

$$\phi \xrightarrow{\text{bias}} \text{attention} \xrightarrow{\text{aggregate}} \text{output} \xrightarrow{W_d} \text{deposit} \xrightarrow{\text{accumulate}} \phi \quad (13)$$

Remark 2 (Causality). *Position i 's deposit $d_i^{(h)}$ depends on the field at positions $0, \dots, i-1$ (through the attention bias) and on the values at positions $0, \dots, i$ (through the attention output). The deposit at position i only affects the field at positions $i+1, i+2, \dots$. This maintains strict causality: no position can influence its own field bias, analogous to how an ant's deposit only affects future ants.*

Remark 3 (Computational cost). *The feedback architecture requires sequential processing over positions (an $O(T)$ loop) because each position's output depends on the field state, which depends on all previous outputs. In parallel mode (training), this can be amortized: the field recurrence is a single scan operation, and the attention computation at each step only requires the pre-computed Q, K, V projections. The overhead per position is a single linear projection W_d (size $d \rightarrow H$) and a scalar multiply-add per head for the field update.*

4 Analysis

4.1 The Field as Modulated State-Space Model

In the non-feedback variant, the pheromone field is a standard linear SSM with fixed dynamics. In the feedback variant, the effective dynamics become input-dependent:

Proposition 3 (Feedback field as nonlinear SSM). *The feedback field can be written as:*

$$\phi_i = \alpha \cdot \phi_{i-1} + f_\theta(\mathbf{x}_{0:i}, \phi_{0:i-1}) \quad (14)$$

where f_θ is the composition of attention (with field-dependent weights) and the deposit projection. This is a nonlinear SSM: the transition function depends on both the input sequence and the state history through the attention mechanism.

This is qualitatively different from linear SSMs (S4, Mamba): the state dynamics are *modulated* by the attention mechanism, which itself is modulated by the state. The field and attention co-evolve.

4.2 Information-Theoretic Interpretation

The three ingredients admit an information-theoretic reading:

- **Erosion** ($\alpha < 1$): Implements an information horizon. The contribution of position j to the field at position i decays as α^{i-j} , creating an exponential forgetting curve. The effective memory span is $\tau_{\text{eff}} = -1/\ln(\alpha) \approx 1/\varepsilon$ for small ε .
- **Injection**: Ensures the field receives fresh signal at every position, preventing collapse to a fixed point. The deposit d_i injects new information proportional to the local content, maintaining exploration of the input space.
- **Amplification**: Creates selective reinforcement. Positions that receive high attention produce outputs that generate deposits that increase future attention to positions with similar patterns. This is a form of *iterated distillation*: the field accumulates a compressed summary of what the attention mechanism has found useful.

The evaporation rate ε controls the balance:

- $\varepsilon \rightarrow 0$ (frozen): infinite memory, field dominated by early deposits, attention locked to initial patterns
- $\varepsilon \rightarrow 1$ (disordered): instant forgetting, field reflects only the most recent deposit, no accumulation
- ε at criticality: memory span matches the task’s relevant context window

4.3 Relationship to Existing Architectures

Architecture	State dim.	Feedback	Attention	Decay
Standard Transformer	—	—	content-based	—
ALiBi	—	—	content + static bias	—
Feedback Transformer	d	nonlinear	content-based	—
RetNet	d	linear	via γ -mask	multiplicative
S4 / Mamba	n	linear	selective (Mamba)	structured A
Linear Attention	$d_k \times d_v$	linear	via kernel	implicit
Stigmergic (input)	H	none	content + field bias	additive
Stigmergic (feedback)	H	nonlinear	content + field bias	additive

Table 2: Comparison of recurrent/attention architectures. H = number of heads, d = model dimension, n = SSM state dimension.

Stigmergic attention decomposes into three familiar components—a scalar recurrence, an additive logit bias, and output feedback—each of which appears independently in prior work. The contribution is their specific combination, motivated by the three-ingredient theory rather than by architectural search.

Additive logit biases. ALiBi [8] adds a fixed, non-learned linear penalty to attention logits based on key–query distance, improving length generalization without positional embeddings. T5 relative position biases [9] learn a per-distance scalar bias. Both are *static*: the bias depends only on position, not on content or state. The pheromone field replaces these fixed biases with a content-dependent recurrence—it is “ALiBi where the slope is a running state driven by what attention has found useful.”

Output feedback. The Feedback Transformer [10] feeds the full d -dimensional output of the top layer back as additional context for the next position’s attention computation. This implements the same principle of output-dependent state, but through a full-dimensional vector used as an extra key–value pair rather than a scalar logit bias. The stigmergic field is a more constrained version: a single scalar per head, which makes the recurrence interpretable and cheap but limits expressivity.

SSM–attention hybrids. Jamba [11], Griffin [12], and RWKV [13] interleave SSM or linear-recurrence layers with attention layers. In all cases the two mechanisms are *decoupled*: the SSM layer processes its input independently, then passes its output to attention (or vice versa). There is no feedback loop between the recurrent state and the attention weights. Mamba-2 [14] shows a formal duality between SSMs and structured attention, but replaces softmax attention with a linear variant rather than augmenting it.

What is novel. The specific wiring—a scalar recurrence whose output additively biases softmax attention logits, with the recurrence input derived from attention output—does not appear in prior work to our knowledge. However, the individual components are not new, and a reviewer could fairly characterise the architecture as “ALiBi with a content-dependent recurrence” or “a 1-d Mamba biasing attention logits.” We do not claim the architecture itself as the primary contribution. Rather, the value lies in the *theoretical framework*: the three-ingredient recipe from ant trail phase transitions provides principled answers to design questions that would otherwise be arbitrary—why scalar (not vector)? why additive (not multiplicative)? why from output (not input)? why uniform decay (not state-dependent)? Without this framework, the non-feedback variant (Section 3.3) looks equally reasonable, and there is no theoretical reason to predict its failure.

5 Experiments

5.1 Setup

Model. A micro-GPT with 2 layers, 4 heads, $d_{\text{model}} = 48$, ReLU activation, RMSNorm, and $\sim 58\text{K}$ parameters. This deliberately small scale allows exhaustive sweeps with multiple seeds.

Task. In-context modular arithmetic: each sequence presents 8 input-output pairs (x_i, y_i) where $y_i = (ax_i + b) \bmod V$ for a random function (a, b) , followed by a query x_q . The model must predict y_q . Vocabulary $V = 16$, sequence length $T = 18$. Training and test function sets are disjoint (200 train / 40 test functions), ensuring evaluation measures out-of-distribution generalization.

Training. AdamW optimizer, learning rate 3×10^{-3} with linear decay, weight decay 0.10, dropout 0.05, batch size 64, 5000 steps. Validation accuracy averaged over last 5 checkpoints (every 50 steps).

5.2 Non-Feedback Field (Negative Result)

We first sweep evaporation rate $\varepsilon \in [0, 0.40]$ with the non-feedback architecture (deposits from input, Equation (7)).

ε	Val. accuracy	$\pm$ std	Seeds
0.00	0.656	0.015	3
0.05	0.652	0.012	3
0.10	0.653	0.018	3
0.15	0.648	0.011	3
0.20	0.674	0.009	3
0.25	0.663	0.011	3
0.30	0.651	0.012	3
0.40	0.722	0.089	3
No field	0.722	—	3

Table 3: Non-feedback (decay) field: no ε value matches the no-field baseline. The field is inert.

As predicted by Proposition 2, the non-feedback field provides no benefit. All ε values produce validation accuracy at or below the no-field baseline. There is no phase transition—the field is a harmless but useless exponentially-weighted positional encoding.

5.3 Feedback Field

We repeat the sweep with the feedback architecture (deposits from output, Equation (12)).

ε	Val. accuracy	$\pm$ std	Seeds
0.00	0.654	0.016	3
0.05	0.715	0.080	3
0.10	0.704	0.069	3
0.15	0.654	0.015	3
0.20	0.662	0.009	3
0.25	0.713	0.086	3
0.30	0.697	0.074	3
0.40	0.647	0.018	3
No field	0.722	0.069	3

Table 4: Feedback field results: deposits computed from attention output. The feedback mode shows higher variance and seed-dependent bimodality (one seed consistently reaches ~ 0.80 – 0.83 while others stay at ~ 0.65), suggestive of a phase transition that the model can reach but does not reliably find.

The feedback results are nuanced. The mean accuracy does not exceed the no-field baseline, but the *variance structure* is qualitatively different from the decay mode. In the decay mode, all seeds cluster tightly around 0.65. In the feedback mode, seed 137 consistently reaches 0.80–0.83 across multiple ε values while other seeds remain at ~ 0.65 . This bimodality is characteristic of systems near a phase transition: the feedback loop creates a basin of attraction that some initializations find and others miss.

Remark 4 (Seed-dependent bimodality). *The high-variance points ($\varepsilon = 0.05, 0.10, 0.25, 0.30$) all show the same pattern: one seed near 0.80, two seeds near 0.65. This is not random noise—it is consistent across ε values for the same seed. The feedback loop appears to create a useful attractor that requires specific initialization conditions to reach, suggesting that the optimization landscape has multiple basins separated by barriers.*

5.4 Length Generalization on State-Tracking Tasks

The modular arithmetic task is fundamentally about random-access pattern matching—the model looks at examples, spots the function, and applies it. Standard attention already excels at this. A more principled evaluation tests the field on tasks requiring *sequential state accumulation*, evaluated on *out-of-distribution sequence lengths*.

We train on sequences of length $T = 34$ ($n = 16$ examples) and test on $T \in \{34, 66, 98, 130\}$, with three tasks:

- **Parity:** given random tokens $x_0, \dots, x_n \in [0, V - 1]$, predict the running parity $(x_0 + \dots + x_i) \bmod 2$ at each position. This is a classic transformer limitation: parity requires aggregating over *all* previous inputs and is not reducible to attending to a single position.
- **Flip-flop:** a register machine where token 0 is a no-op and any other token k sets the state to k . The model must predict the current state—i.e., the last non-zero write, which may be arbitrarily far back. This tests long-range memory retention.
- **Cumulative sum:** predict $(x_0 + \dots + x_i) \bmod V$ at each position. This requires precise modular arithmetic, not just binary state.

All tasks are trivially solvable with a running state variable (a single register updated at each step) and all models achieve 1.000 accuracy at the training length $T = 34$. The question is whether this capability *generalizes* to unseen lengths.

	$T = 34$ (train)	$T = 66$ (OOD)	$T = 98$ (OOD)	$T = 130$ (OOD)
Parity				
No field	1.000	0.542 $\pm$.021	0.512 $\pm$.010	0.514 $\pm$.013
Feedback $\varepsilon=0.05$	1.000	0.615 $\pm$.065	0.526 $\pm$.029	0.554 $\pm$.058
Feedback $\varepsilon=0.10$	1.000	0.540 $\pm$.045	0.536 $\pm$.059	0.525 $\pm$.022
Feedback $\varepsilon=0.20$	1.000	0.540 $\pm$.045	0.535 $\pm$.049	0.521 $\pm$.013
Flip-flop				
No field	1.000	0.524 $\pm$.007	0.516 $\pm$.002	0.521 $\pm$.003
Feedback $\varepsilon=0.05$	1.000	0.557 $\pm$.011	0.518 $\pm$.004	0.524 $\pm$.009
Cumulative sum				
No field	1.000	0.098 $\pm$.013	0.072 $\pm$.012	0.078 $\pm$.014
Feedback $\varepsilon=0.05$	1.000	0.083 $\pm$.012	0.076 $\pm$.027	0.054 $\pm$.008

Table 5: Length generalization: train on $T = 34$, test on longer sequences. Random baseline is 0.5 for parity, $1/V \approx 0.063$ for cumsum, $\sim 1/V$ for flip-flop. Bold marks the best result per row where it exceeds no-field by $>1\sigma$. Three seeds per condition.

The results reveal a clear task-dependent pattern:

Parity (positive result). The feedback field at $\varepsilon = 0.05$ improves OOD accuracy from $0.542 \rightarrow 0.615$ at $T = 66$ and from $0.514 \rightarrow 0.554$ at $T = 130$. The baseline drops to random (0.50) immediately outside the training distribution, while the feedback field retains partial generalization. This is the first clean positive result: the scalar recurrence carries parity state—literally a running XOR—beyond what position-tied attention patterns can represent.

Flip-flop (modest signal). Feedback $\varepsilon = 0.05$ shows a consistent improvement at $T = 66$ (0.557 vs 0.524) but the effect fades at longer lengths. The flip-flop task requires *selective over-writing*—the field must learn to replace its state when a non-zero token appears and maintain it through no-ops. A single scalar recurrence with fixed decay rate is a poor fit: the decay is unconditional, so the state fades even when it should persist.

Cumulative sum (negative result). Both modes collapse to near-random at $T = 66$. Cumulative sum mod $V = 16$ requires precise 4-bit modular arithmetic—a scalar field with continuous values cannot faithfully encode a discrete counter. The task demands *multi-bit state* that exceeds the capacity of H scalar channels.

Why parity works and cumsum does not. The key distinction is *state complexity*. Parity has 1 bit of state (even/odd). A single scalar field head can encode this as the sign of the field value, and the sign is preserved under the exponential decay (positive stays positive, negative stays negative). Cumulative sum mod 16 has $\lceil \log_2 16 \rceil = 4$ bits of state. To encode this, the field would need to maintain 4 independent channels that evolve without interference—exactly what the $H = 4$ heads could in principle provide, but the learned positional embeddings (which do not generalize to unseen positions) destroy the field’s contribution at OOD lengths.

This suggests a clear scaling law for stigmergic attention: the feedback field can track $O(H)$ bits of state, and should outperform standard attention on tasks whose state complexity is $\leq H$ bits, evaluated at lengths beyond the training distribution.

6 Discussion

6.1 What the Bug Revealed

Our investigation began with a serendipitous bug: an inverted decay exponent that caused the decay matrix to *amplify* old deposits ($D_{ij} = \alpha^{j-i}$ with $j < i$, giving $D_{ij} > 1$) rather than decay them. This buggy “amplify” mode produced a striking phase transition—but one driven by exponential growth rather than stigmergic dynamics. Fixing the bug eliminated the phase transition entirely, revealing that the correct-decay field was inert.

The ant paper provided the explanation: the architecture had injection (content-dependent deposits) and erosion (uniform decay) but lacked *amplification* (positive feedback). The buggy amplification accidentally provided a crude substitute—not through the stigmergic mechanism, but through raw exponential growth of the field values.

6.2 The Feedback Hypothesis

The feedback architecture (Definition 4) is the minimal change that closes the loop. It adds no parameters (same W_d projection) and changes only the *source* of deposits: from the attention input $\mathbf{x}_i$ to the attention output $\mathbf{o}_i$.

This creates qualitatively different dynamics:

- **Self-reinforcement:** positions that receive high attention produce outputs that generate deposits, which increase future attention to those positions.
- **Trail formation:** repeatedly useful patterns create persistent field structures that guide future attention.
- **Competition:** uniform erosion ensures that only actively reinforced structures persist—abandoned trails fade.

6.3 Limitations

- **Scale:** All experiments use a $\sim 58K$ parameter model on a single synthetic task. The feedback mechanism’s value at transformer scale is unknown.
- **Sequential bottleneck:** The feedback loop requires position-by-position processing during training, losing the parallelism advantage of standard attention. For long sequences, this may be prohibitive without approximation.
- **Task specificity:** In-context modular arithmetic is a controlled but narrow evaluation. Natural language tasks may not benefit from the same mechanism.

7 Research Directions

7.1 Finding the Right Task

The modular arithmetic task may be the wrong testbed. It is fundamentally about *algebraic structure*—recognizing (a, b) from examples and applying a formula. Attention is already good at this: match the query x against example x ’s, retrieve the corresponding y , infer the pattern.

A stigmergic field should shine on tasks where **sequential accumulation matters**—where the answer depends not on retrieving a single example but on building up a representation *across* the sequence. Three candidate task families:

- **Counting and frequency estimation:** “Which token appeared most often?” requires accumulating evidence across all positions, not just attending to one. The field naturally accumulates; attention does not.
- **Order-dependent state tracking:** “Follow these instructions in sequence”—where each step modifies a state. A maze walk, a state machine, a sequence of edits. The field’s recurrence is literally a running state.
- **Multi-hop reasoning with fading evidence:** “A is north of B, B is north of C, where is A relative to C?” spread across many distractor tokens. The field can maintain a decaying trace of B’s location while processing distractors, bridging the gap without needing to attend all the way back.

The common thread: tasks where a **running summary** is more useful than **random-access retrieval**. Standard attention excels at random access (query any position). The field adds a complementary channel—a lossy, compressed, sequential summary. If the task does not need that, the field is dead weight.

The modular arithmetic task is almost pure random access: look at the examples, spot the pattern, apply it. No wonder the field does not help.

A concrete proposal: a **state-tracking task** such as “given a sequence of +1 and −1 tokens, is the running sum positive or negative at the query position?” This is trivially easy with a running state and provably hard for bounded-depth transformers. If the feedback field does not help *there*, it is dead. If it does, we have found the right problem class.

7.2 Alternative Architectural Framings

The feedback loop implements a general pattern:

state → modulate processing → observe result → update state → repeat

This pattern appears in several architectural families beyond attention, suggesting alternative carriers for the stigmergic mechanism.

Energy-based models. The field can be interpreted as an energy landscape that attention navigates. Deposits lower energy at useful positions, erosion raises it everywhere, and attention follows the energy gradient. This maps to Hopfield networks—modern Hopfield layers [6] already connect energy models to attention. The feedback field could be an *asymmetric* Hopfield energy that evolves causally.

Predictive coding. Under the predictive coding hypothesis, each layer maintains a prediction of its input, computes prediction errors, and updates. The field serves as the prediction (running estimate of what is important), the attention output is the observation, and the deposit is the prediction error. This framing naturally explains why deposits should come from the *output* not the *input*—the deposit is the *surprise*, not the stimulus.

Recurrent attention. Universal Transformers and PonderNet iterate the same attention layer multiple times. The feedback field is a lightweight version: instead of rerunning full attention, it carries forward a scalar summary per head. Much cheaper, but the same principle of iterative refinement.

Diffusion models. In diffusion models, a noisy signal is iteratively refined by predicting and removing noise. The pheromone field could be seen as a “structure estimate” that gets refined position by position. However, diffusion operates in continuous space with a learned score function, while our field is a scalar per head. The analogy is suggestive but the mapping is loose.

We note that the hypothesis is about the *feedback mechanism*, not the specific carrier. If positive feedback between a decaying state and a processing layer produces useful dynamics, the principle should transfer across architectures. We therefore resist switching carriers until the mechanism is validated on a suitable task.

7.3 Theoretical Limitations of Transformers

The strongest motivation for stigmergic attention comes from known theoretical limitations of transformers that a feedback field directly addresses.

State tracking. Transformers are provably bad at simulating automata or tracking state over long sequences [7]. They cannot count, cannot maintain a running tally, cannot simulate a state machine beyond a few steps. The field IS a running state—a scalar recurrence per head. It does not give a full state machine, but it provides H scalar channels of persistent state that attention can read and write. That is exactly what is missing.

Length generalization. Transformers trained on length N fail at length $N + k$. This is because attention patterns are tied to absolute or relative positions. The field’s decay creates a *scale-invariant* memory—the exponential decay α^{i-j} depends only on distance, not absolute position. A feedback field trained on short sequences might generalize to longer ones because the dynamics are position-independent.

Compositional generalization. Multi-step reasoning requires chaining intermediate results. Attention can only do this by allocating positions as “scratchpad” (chain-of-thought). The field provides an implicit scratchpad—each position’s output modifies the field, carrying forward a compressed result without consuming a token position.

The clearest theoretical pitch: **transformers are memoryless between positions**—each position’s computation is independent given the input. The field adds causal memory. The feedback loop makes that memory *adaptive* (content-dependent, not just positional). This is the one thing attention provably cannot do and the field provably can.

8 Other Perspectives: Multi-Scale and Hierarchical Fields

The single-scale, single-layer field presented so far is the minimal instantiation of stigmergic dynamics. But the three-ingredient recipe is a *local* constraint—it specifies the conditions for edge-of-chaos dynamics within a single field. This raises a natural question: what happens when multiple fields interact across timescales or across layers? We argue that the uncoupling requirements (injection and erosion uncorrelated with the field) need only hold *within* a scale, not *between* scales. Signals arriving from a different timescale or a different layer are, from the perspective of the receiving field, external inputs—structurally analogous to injection, not to self-correlated feedback.

8.1 Multi-Scale Fields

Instead of a single field per head, assign each head a *different* evaporation rate, or equip each head with multiple fields at distinct timescales. In the simplest version, head h operates at

retention rate α_h , and the collection of heads spans a range from fast (small α , local pattern tracking) to slow (large α , global structure).

A richer variant couples a fast field and a slow field per head:

$$\phi_i^{\text{fast}} = \alpha_f \cdot \phi_{i-1}^{\text{fast}} + d_i \quad (15)$$

$$\phi_i^{\text{slow}} = \alpha_s \cdot \phi_{i-1}^{\text{slow}} + h(\phi_i^{\text{fast}}) \quad (16)$$

where $\alpha_s \gg \alpha_f$ and h is a learned scalar projection. The fast field tracks recent attention patterns via output deposits as before. The slow field accumulates a summary of *which fast-field patterns persist*—it is a distillation of the distillation. Both fields contribute additive biases to the attention logits:

$$\ell_{ij}^{(h)} = \frac{\mathbf{q}_i^{(h)\top} \mathbf{k}_j^{(h)}}{\sqrt{d_h}} + \phi_j^{\text{fast}} + \phi_j^{\text{slow}} \quad (17)$$

The biological analogue is direct: social insects use multiple pheromone species with different volatilities. Trail pheromone (fast) marks recent paths and evaporates in minutes; colony pheromone (slow) marks established routes and persists for hours. The slow pheromone does not respond to individual ant decisions—it integrates over the fast pheromone’s history.

Why the three-ingredient recipe still holds. Consider the slow field in isolation. Its deposit source is $h(\phi^{\text{fast}})$, which depends on the fast field but *not* on ϕ^{slow} itself. From the slow field’s perspective, this is injection: an external signal uncorrelated with its own state. Its erosion ($\alpha_s < 1$) is uniform. Its amplification comes indirectly: the slow field biases attention, which shapes the output, which feeds the fast field, which eventually feeds the slow field—but this loop passes through the fast field’s own dynamics, maintaining the separation of scales. The uncoupling constraints hold *within each scale* even though the scales are coupled *between* each other.

This is analogous to how weather (slow) affects pheromone evaporation rates without violating the local dynamics of trail formation: the weather is external to the ant–pheromone system even though it modulates it.

8.2 Cross-Layer Fields

In a multi-layer transformer, each layer currently maintains an independent field with no memory of previous layers. A natural extension passes field state *vertically*. For a network with layers $L = 1, \dots, \mathcal{L}$, the field at layer L evolves as:

$$\phi_i^{(L)} = \alpha^{(L)} \cdot \phi_{i-1}^{(L)} + d_i^{(L)}, \quad d_i^{(L)} = W_d^{(L)} \mathbf{o}_i^{(L)} \quad (18)$$

with the cross-layer initial condition:

$$\phi_0^{(L+1)} = g^{(L)}(\phi_T^{(L)}) \quad (19)$$

where $\phi_T^{(L)}$ is the terminal field state of layer L and $g^{(L)}$ is a learned scalar projection. Layer $L + 1$ inherits a landscape shaped by layer L ’s attention patterns and refines it with its own deposits.

This creates **hierarchical feature accumulation**: layer 1’s field summarises token-level patterns (which tokens were attended to), layer 2’s field summarises layer 1’s abstractions (which *patterns of attention* recurred), and so on. Each layer refines the previous layer’s trail map at a higher level of abstraction. Unrolling the full cross-layer field at layer $L + 1$:

$$\phi_i^{(L+1)} = (\alpha^{(L+1)})^i \cdot g^{(L)}(\phi_T^{(L)}) + \sum_{j=0}^i (\alpha^{(L+1)})^{i-j} \cdot d_j^{(L+1)} \quad (20)$$

The first term is the inherited landscape, decaying as the new layer processes the sequence; the second term is the new layer’s own deposits. The relative influence of inheritance vs. fresh deposits is controlled by the retention rate and sequence length.

The same within-scale uncoupling argument applies. Layer $L + 1$ ’s field receives $g(\phi_T^{(L)})$ as an initial condition—a one-time injection from an external source. The field’s own recurrence, deposits, and erosion then proceed independently. The cross-layer signal is structurally identical to an initial deposit: external, uncorrelated with the receiving field’s future state.

Combining multi-scale and cross-layer. The two extensions compose naturally. Each layer L maintains fast and slow fields; the slow field’s terminal state seeds the next layer’s slow field:

$$\phi_i^{(L,f)} = \alpha_f^{(L)} \cdot \phi_{i-1}^{(L,f)} + d_i^{(L)} \quad (21)$$

$$\phi_i^{(L,s)} = \alpha_s^{(L)} \cdot \phi_{i-1}^{(L,s)} + h^{(L)}(\phi_i^{(L,f)}) \quad (22)$$

$$\phi_0^{(L+1,s)} = g^{(L)}(\phi_T^{(L,s)}) \quad (23)$$

with the attention logit at layer L :

$$\ell_{ij}^{(L,h)} = \frac{\mathbf{q}_i^{(L,h)\top} \mathbf{k}_j^{(L,h)}}{\sqrt{d_h}} + \phi_j^{(L,f)} + \phi_j^{(L,s)} \quad (24)$$

This gives the network a hierarchy along two axes: *temporal* (fast vs. slow within a layer) and *representational* (shallow vs. deep across layers). The result is a lattice of interacting fields, each governed locally by the three-ingredient recipe, coupled only through external channels that preserve the uncoupling constraints.

8.3 Adaptive Criticality

A more speculative extension treats the evaporation rate itself as a field. A *volatility field* ψ_i modulates the primary field’s retention:

$$\phi_i = \alpha(\psi_i) \cdot \phi_{i-1} + d_i \quad (25)$$

$$\psi_i = \alpha_\psi \cdot \psi_{i-1} + d_i^{(\psi)} \quad (26)$$

where $\alpha(\psi) = \sigma(\psi)$ maps the volatility field to a retention rate via a sigmoid, and $\alpha_\psi \approx 1$ ensures ψ evolves on a much slower timescale than ϕ .

This directly raises the question of whether the three-ingredient recipe tolerates inter-scale coupling at all. The lattice ant model tested state-dependent erosion and found that it destroys edge-of-chaos dynamics in every case—but those tests modulated erosion as a function of the field’s *own* value at the *same* position and timescale. The multi-scale and cross-layer extensions proposed above involve a different structure: modulation by a signal that is external to the receiving field’s dynamics. We state this distinction as a hypothesis.

Hypothesis (Scale-local uncoupling). *The three-ingredient conditions for edge-of-chaos dynamics—amplification, uncorrelated injection, and uncorrelated erosion—are scale-local: they must hold within a single field’s characteristic timescale τ_{eff} , but may be violated by signals arriving from a sufficiently separated timescale.*

Why this might be correct. The argument is an adiabatic approximation. If the volatility field ψ varies on a timescale $\tau_\psi \gg \tau_\phi$, then over any window of length $\sim \tau_\phi$ the retention rate $\alpha(\psi)$ is effectively constant. The fast field experiences locally uniform erosion at every instant,

which is exactly what the recipe requires. The slow modulation shifts the fast field’s operating point over time but does not introduce within-scale correlation between the erosion rate and the field value. This is the same reasoning that allows thermodynamics to treat temperature as a constant in chemical kinetics even though temperature is itself changing: the reaction timescale is fast relative to the thermal timescale. In biological terms, weather modulates pheromone evaporation rates without violating the local trail-formation dynamics—rain is external to the ant-pheromone system even though it changes the evaporation rate everywhere.

The same logic extends to cross-layer fields (Section 8.2): the inherited initial condition $g(\phi_T^{(L)})$ is a one-time injection from an external source. And to multi-scale fields (Section 8.1): the slow field’s deposit $h(\phi^{\text{fast}})$ depends on the fast field, not on the slow field’s own state. In each case, the inter-scale signal is structurally external to the receiving field.

Why this might be wrong. Three concerns deserve scrutiny:

1. **The ant paper did not test this.** All seven failed mechanisms in [1] involved within-scale state dependence. Slowly varying external modulation was not tested. The hypothesis is an extrapolation, not an established result.
2. **Bifurcation crossings.** The adiabatic approximation fails when the slow variable passes through a bifurcation point of the fast system. If ψ drifts through the critical evaporation rate ε^* , the fast field undergoes an abrupt phase transition rather than a smooth parameter change. The fast dynamics cannot track a discontinuity adiabatically. In the neural setting this might be benign (the network learns to avoid the critical point) or catastrophic (training becomes unstable near the transition). This is an empirical question.
3. **“Sufficiently separated” is vague.** How large must τ_ψ/τ_ϕ be? The adiabatic approximation is a continuum of accuracy, not a binary condition. At moderate separation the fast field experiences erosion that is *approximately* uniform but not exactly so—enough correlation to degrade edge-of-chaos dynamics without obviously breaking them. The required separation ratio is likely task-dependent and must be determined empirically.

A testable prediction. The lattice ant model provides a direct testbed. Modulate the evaporation rate λ sinusoidally at period P and measure the edge-of-chaos order parameter as a function of P/τ_ϕ . The hypothesis predicts that edge-of-chaos dynamics survive when $P \gg \tau_\phi$ and collapse when $P \sim \tau_\phi$. The crossover point would quantify the required timescale separation. If confirmed, this would establish scale-local uncoupling as a general principle for stigmergic systems; if falsified, it would constrain the multi-scale and adaptive extensions to purely static (non-state-dependent) parameter choices.

8.4 Experimental Feasibility at Current Scale

The micro-GPT used in Section 5 (2 layers, 4 heads, $d_{\text{model}} = 48$, $T = 18$) is sufficient to test some of the proposed extensions but not all. We summarise the feasibility constraints to guide the experimental programme.

Multi-scale fields (Section 8.1). *Feasible.* With 4 heads, the simplest test assigns different evaporation rates per head—e.g., 2 fast heads (ε_f) and 2 slow heads (ε_s)—and compares against uniform ε across all heads. This requires no architectural changes beyond making ε a per-head parameter. The coupled fast-slow variant (Equations (15) and (16)) adds one scalar projection h per head, a negligible parameter increase.

However, the short sequence length ($T = 18$) constrains the range of useful timescales. A slow field with $\alpha_s = 0.99$ has effective memory $\tau_{\text{eff}} = -1/\ln(0.99) \approx 100$, far exceeding the

sequence—it is effectively frozen, accumulating all deposits with near-zero decay. A fast field with $\alpha_f = 0.90$ has $\tau_{\text{eff}} \approx 10$, covering most of the sequence. Meaningful separation requires:

$$1 \ll \tau_{\text{eff}}^{\text{fast}} < T \ll \tau_{\text{eff}}^{\text{slow}} \quad (27)$$

At $T = 18$ there is limited dynamic range for the fast field to operate in a non-trivial regime while leaving room for a separated slow field.

Cross-layer fields (Section 8.2). *Feasible at minimum depth.* Two layers provide exactly one cross-layer handoff: layer 1’s terminal field state $\phi_T^{(1)}$ seeds layer 2’s initial condition via Equation (19). This is the minimal test of whether field inheritance helps at all. It cannot show whether the effect compounds over depth. A 4-layer model provides three cross-layer handoffs—enough to observe whether hierarchical accumulation compounds or saturates.

Adaptive criticality (Section 8.3). *Not feasible at current scale.* The volatility field ψ must vary meaningfully *during* the sequence for the adaptive mechanism to differ from a static ε . At $T = 18$, a slowly varying ψ with $\alpha_\psi \approx 1$ is effectively constant across the sequence—indistinguishable from a learned fixed ε . Testing adaptive criticality requires sequences long enough for the slow field to traverse a non-trivial trajectory, suggesting $T \geq 128$.

Recommended progression. The experimental programme proceeds in three stages:

1. **Validate the base mechanism** on a task that requires sequential accumulation (state tracking: running sum of ± 1 tokens, as proposed in Section 7.1). If the single-scale feedback field does not help here, the hierarchical extensions are moot.
2. **Test multi-scale and cross-layer** on the same task at $T = 18$. Per-head evaporation rates and one cross-layer handoff are cheap additions. Compare: (a) uniform ε feedback, (b) per-head ε feedback, (c) cross-layer feedback, (d) both.
3. **Scale up** to $T \in [128, 256]$ with a 4-layer model ($d_{\text{model}} \geq 96$) to test adaptive criticality and deep cross-layer accumulation. At this scale, the timescale separation constraint (Equation (27)) admits a meaningful range, and the bifurcation-crossing concern becomes empirically testable.

9 Conclusion

We presented stigmergic attention, an architecture that augments transformer attention with a decaying pheromone field. Guided by the phase transition theory of lattice ant models, we identified that the critical ingredient for non-trivial dynamics is *positive feedback*—computing deposits from attention output, not input.

The non-feedback variant is provably inert: a causally-weighted moving average that adds nothing beyond positional information. The feedback variant closes the loop between field and attention, creating a nonlinear state-space model coupled to the attention mechanism.

The length generalization experiments (Section 5.4) provide the clearest evidence for the mechanism. On parity—a canonical state-tracking task—the feedback field improves out-of-distribution accuracy from $0.54 \rightarrow 0.62$ at $2\times$ the training length and from $0.51 \rightarrow 0.55$ at $4\times$, while all models achieve 1.000 at the training length. The effect is task-specific: it appears on parity (1-bit state, within the field’s capacity) but not on cumulative sum mod 16 (4-bit state, exceeding scalar capacity). This pattern suggests a *state-complexity scaling law*: the field should help when the task’s state fits in $O(H)$ scalar channels.

The results are modest in absolute terms but directionally clear: the feedback field provides a form of intrinsic memory that standard attention lacks, and this memory transfers to unseen sequence lengths because the exponential decay α^{i-j} depends on distance, not absolute position. The mechanism maps cleanly to existing architectural concepts—it is simultaneously a content-dependent ALiBi, a minimal SSM coupled to attention, and a neural instantiation of the ant trail recipe.

A fundamental limitation of the scalar-bias architecture is its expressivity: a single scalar per head can bias *how much* attention a position receives, but cannot modulate *what kind* of matching occurs. The field says “attend more here” but not “attend in this direction.” Real pheromone trails encode *gradients*—ants follow concentration differentials, not absolute levels. The neural analogue would be a field that modulates the geometry of the attention space itself—producing vectors that reshape keys or queries rather than scalars that nudge logits. This suggests a successor architecture where the field is a co-equal routing mechanism, not an additive decoration, decomposing attention into structured content matching and experiential trail following. This direction is explored in a companion paper.

References

- [1] Pheromone trail formation at the edge of chaos: Phase diagrams of a lattice ant model. *Research paper*, 2024.
- [2] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *NeurIPS*, 2017.
- [3] A. Gu, K. Goel, and C. Ré. Efficiently modeling long sequences with structured state spaces. In *ICLR*, 2022.
- [4] A. Gu and T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752*, 2023.
- [5] Y. Sun, L. Dong, S. Huang, S. Ma, Y. Xia, J. Xue, J. Wang, and F. Wei. Retentive network: A successor to transformer for large language models. *arXiv:2307.08621*, 2023.
- [6] H. Ramsauer, B. Schäfl, J. Lehner, P. Seidl, M. Widrich, T. Adler, L. Gruber, M. Holzleitner, M. Pavlović, G. K. Sandve, V. Greiff, D. Kreil, M. Kopp, G. Klambauer, J. Brandstetter, and S. Hochreiter. Hopfield networks is all you need. In *ICLR*, 2021.
- [7] W. Merrill and A. Sabharwal. The parallelism tradeoff: Limitations of log-precision transformers. *Transactions of the Association for Computational Linguistics*, 11:531–545, 2023.
- [8] O. Press, N. Smith, and M. Lewis. Train short, test long: Attention with linear biases enables input length extrapolation. In *ICLR*, 2022.
- [9] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *JMLR*, 21(140):1–67, 2020.
- [10] A. Fan, T. Lavril, E. Grave, A. Joulin, and S. Sukhbaatar. Addressing some limitations of transformers with feedback memory. *arXiv:2002.09402*, 2021.
- [11] O. Lieber, B. Lenz, H. Bata, G. Cohen, J. Osin, I. Dalmedigos, E. Safahi, S. Meirom, Y. Belinkov, S. Shalev-Shwartz, and O. Shoham. Jamba: A hybrid transformer-mamba language model. *arXiv:2403.19887*, 2024.

- [12] S. De, S. L. Smith, A. Fernando, A. Botev, G. Cristian-Muraru, A. Gu, R. Harber, L. Hazan, Y.-H. Hsu, et al. Griffin: Mixing gated linear recurrences with local attention for efficient language models. *arXiv:2402.19427*, 2024.
- [13] B. Peng, E. Alcaide, Q. Anthony, A. Albalak, S. Arcadinho, H. Cao, X. Cheng, M. Chung, et al. RWKV: Reinventing RNNs for the transformer era. In *EMNLP Findings*, 2023.
- [14] T. Dao and A. Gu. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In *ICML*, 2024.